# A VLM-Based Approach for 2D Mask Grouping Across Views

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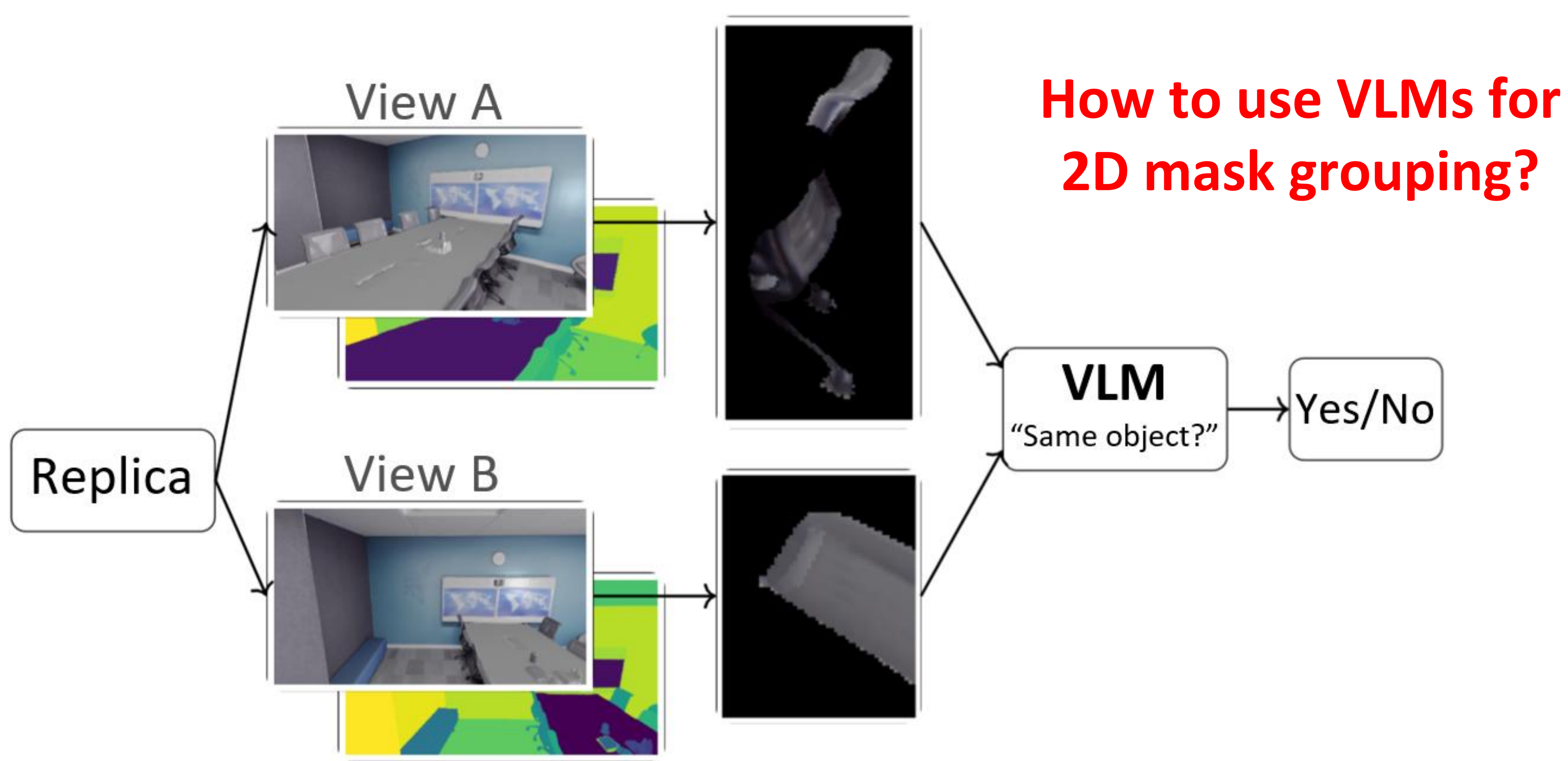
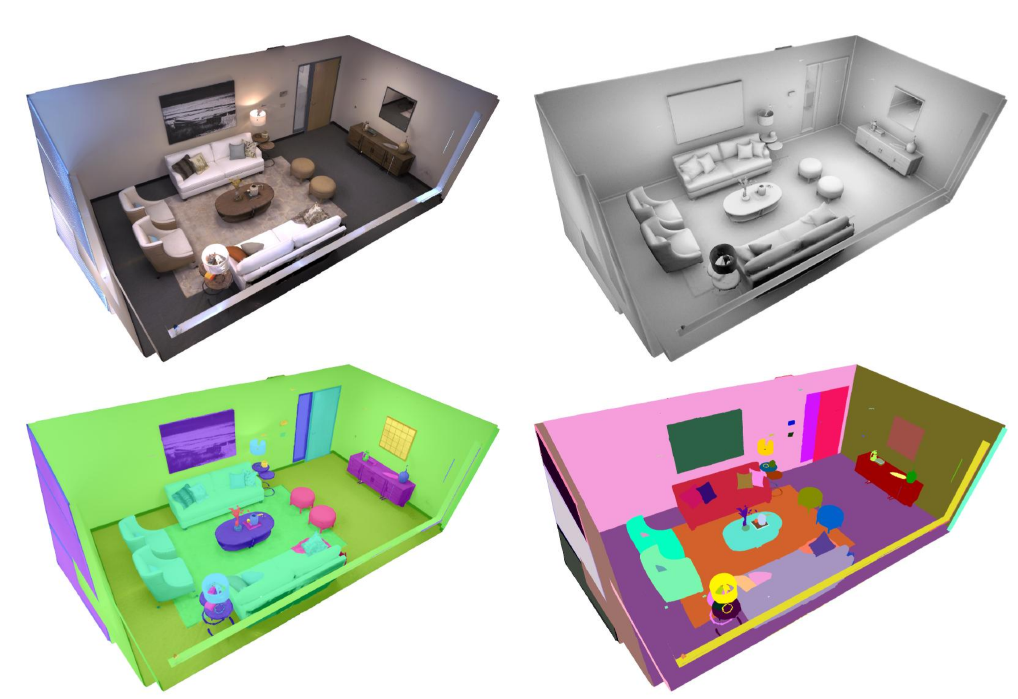
## Our Contribution

- VLM-compatible formulation of multi-view 2D mask grouping
- Baseline evaluation of Qwen3-VL on this task
- Strategy for fine-tuning Qwen3-VL with LoRA, shown to increase consistency without updating base weights




## Replica Dataset

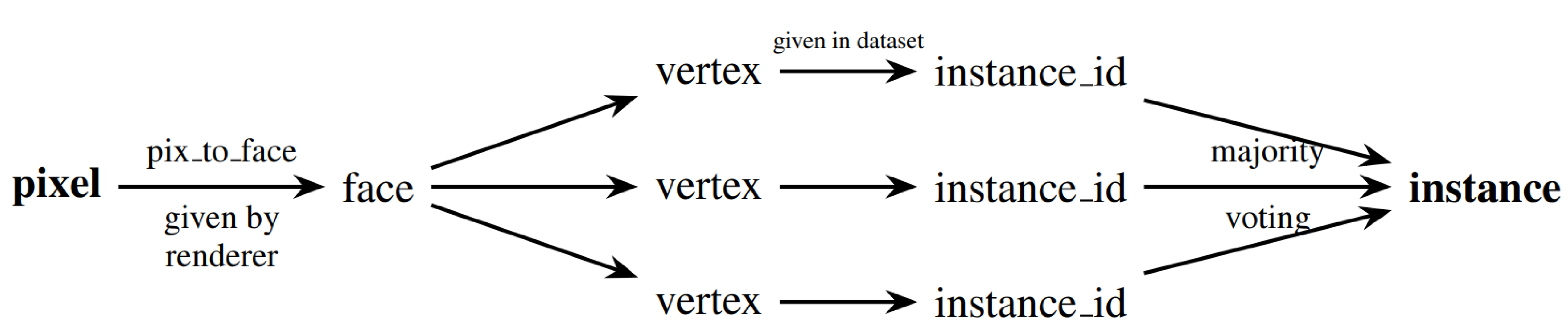


High quality 3D mesh indoor data

- 6 training scenes: 4 offices, 2 rooms
- 2 evaluation scenes: 1 office and 1 room

## Generating Instance Masks from 3D Scenes

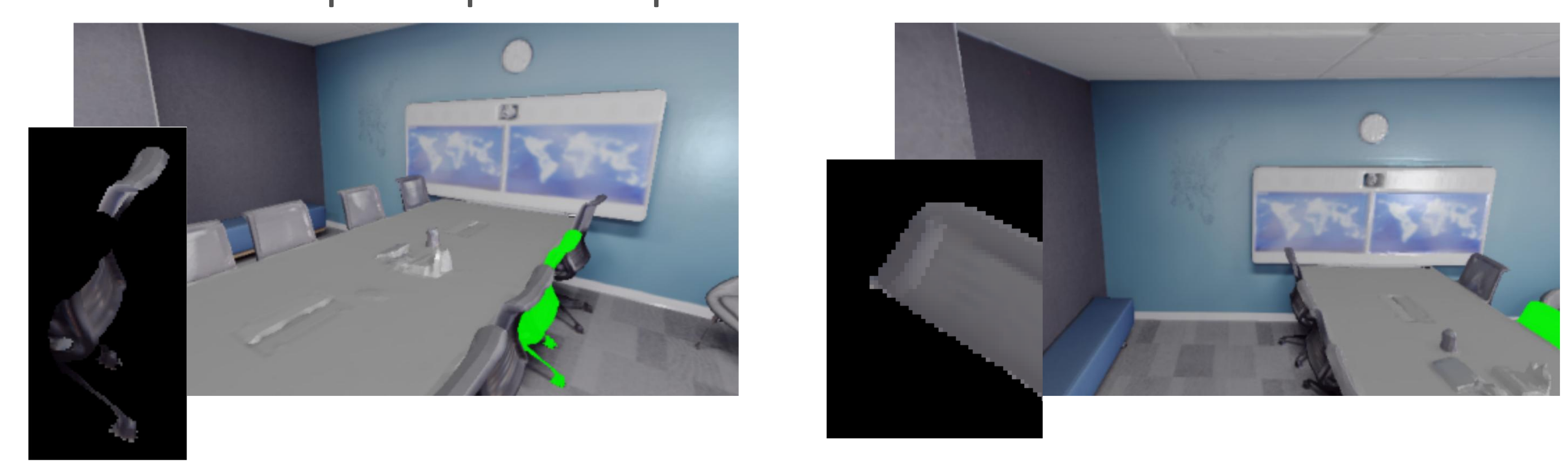
Render 2D views from 3D scenes, then get instance masks by voting, using vertex annotations from Replica




## Constructing Instance Pairs

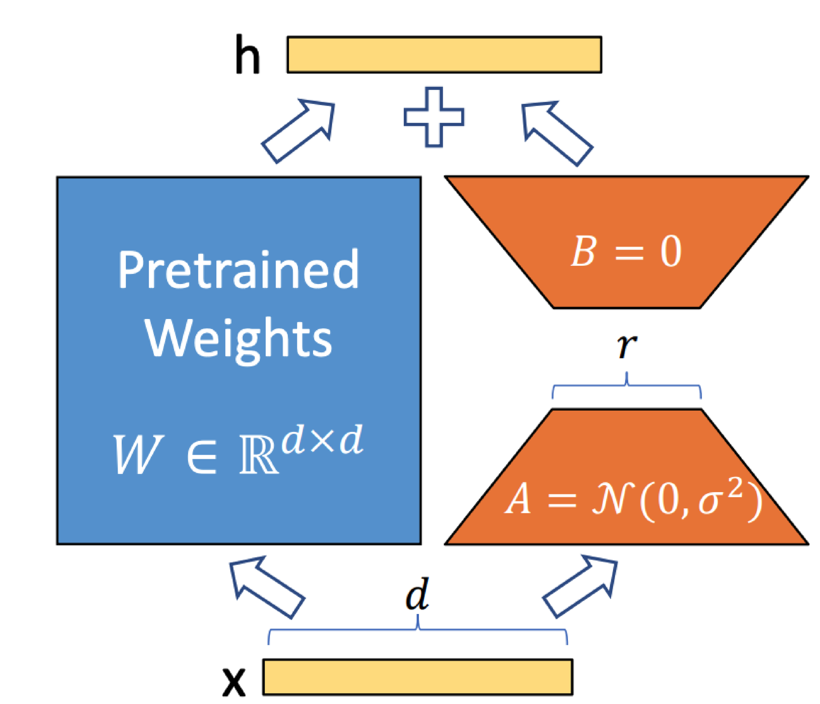
- Sample two views randomly
- Shared instance IDs defines positive pair, else negative
- Crop around instance and add padding

Example of positive pair: two views of the same chair



## LoRA Fine-Tuning

- Freeze original model weights
- Inject low-rank matrix updates
- Fine-tuned Qwen3-VL comparing it to its base version with original weights




## Results

- Results suggest fine-tuning helps internalize task-specific cues
- Qwen consistently outputs yes or no answers
- Typical failure modes include occlusions, symmetry, repetitiveness

Performance of Base vs Fine-Tuned model

| Scene | Model | Acc. | Prec. | Rec. | Amb. |
|---|---|---|---|---|---|
| office4 | Base | 0.74 | 0.67 | 0.88 | 0.00 |
| office4 | LoRA | 0.82 | 0.76 | 0.92 | 0.00 |
| room2 | Base | 0.80 | 0.76 | 0.91 | 0.00 |
| room2 | LoRA | 0.92 | 0.91 | 0.94 | 0.00 |